

PHENIX Preliminary Data Request

Low- p_T Direct Photons

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PWG: Heavy Ion, Photon, PLHF

What Observable: low- p_T direct photon production using the external conversion method; R_γ , direct photon invariant yield, and integrated direct photon yield

System, Energy, Run: Run-12 Cu+Au at $\sqrt{s_{NN}} = 200$ GeV

Last PWG Presentation Link:

<https://phenix-intra.sdcc.bnl.gov/cdsagenda/fullAgenda.php?ida=a2626>

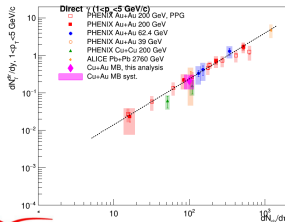
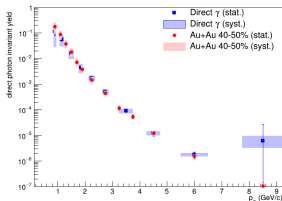
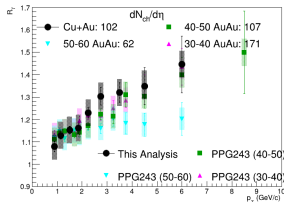
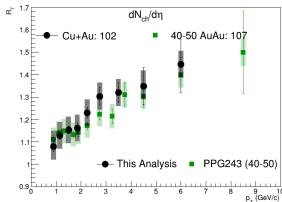
Analysis Note: an1525

Analysis Note Web Link:

https://phenix-intra.sdcc.bnl.gov/phenix/WWW/p/draft/tongzhonguo/Direct_photon_production_Cu_Au_v1.pdf

Plots requested for preliminary status

- R_γ in Cu+Au minimum bias compared with PHENIX Au+Au 40–50%.
- R_γ in Cu+Au minimum bias compared with PHENIX Au+Au 50–60%, 40–50%, and 30–40%.
- Direct photon invariant yield in Cu+Au minimum bias compared with PHENIX Au+Au 40–50%.
- Integrated direct photon yield $dN_\gamma^{\text{dir}}/dy$ for $1 < p_T < 5 \text{ GeV}/c$ versus $dN_{\text{ch}}/d\eta$.



Basic Information & Provenance

Data set:

Run-12 Cu+Au at $\sqrt{s_{NN}} = 200$ GeV, minimum-bias data sample

Data provenance and reconstruction:

CNT, DST_EVE, pro99

Analysis code location:

offline/AnalysisTrain/DileptonAnalysis/
offline/AnalysisTrain/pat/macro/Run_DileptonAnalysis_CuAu.C

Please list all recalibrators used in this analysis:

BeamOffset, MomScale, EMCal

Were any of these not in the MasterRecalibrator?

No

Basic analysis cuts used: next slide

Basic Analysis Cuts

The nominal cut definitions are taken from the analysis note. The main selections are:

- ▶ Event selection: BBCLL1(> 1 tubes)_narrowvtx trigger and $|z_{\text{vtx}}| < 10$ cm.
- ▶ Charged tracks: quality = 31|51|63, $|zed| < 75$ cm, and $p_T > 0.3$ GeV/ c .
- ▶ Electron ID: RICH requirement $n_0 \geq 2$, EMCal matching $|emcdz| < 20$ cm and $|emcd\varphi| < 0.05$ rad, and $E/p > 0.6$ requirement.
- ▶ EMCal photon candidates: cluster E_{core} threshold, shower-shape requirement $\chi^2 < 3$, EMCal dead maps, and an electron/positron veto.
- ▶ Conversion pairs: external conversion-pair cuts based on variables such as $dzed < 4$ cm, $1 < r_{\text{conv}} < 29$ cm.

The systematic study repeats the extraction with controlled variations of these selections, including conversion-pair cuts, electron ID, electron-track momentum threshold, photon threshold, shower shape, energy scale, and π^0 peak extraction.

What's the observable

What observable is analyzed?

- Low- p_T direct photons in Cu+Au minbias at $\sqrt{s_{NN}} = 200$ GeV
- Direct photon double ratio R_γ
- Direct photon invariant yield
- Integrated direct photon yield $dN_\gamma^{\text{dir}}/dy$ for $1 < p_T < 5$ GeV/c

What will be plotted?

- R_γ in 093 Cu+Au compared to PHENIX Au+Au 40–50% from PPG243
- R_γ in 093 Cu+Au compared to PHENIX Au+Au 50–60%, 40–50%, and 30–40% from PPG243
- Direct photon invariant yield in Cu+Au minimum bias compared to PHENIX Au+Au 40–50%
- Integrated direct photon yield $dN_\gamma^{\text{dir}}/dy$ vs. $dN_{\text{ch}}/d\eta$ compared to PHENIX and ALICE measurements

What's new about this result?

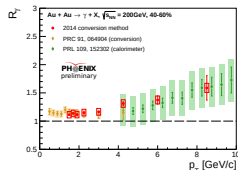
- First low- p_T direct photon result in Run-12 Cu+Au collisions

What is the question that this new result will answer?

- Whether Cu+Au direct photon production follows the same trend with $dN_{\text{ch}}/d\eta$ as Au+Au, Cu+Cu, and Pb+Pb systems

Outline of the analysis method

- If you use an established analysis method that was previously used in PHENIX published paper(s), please list the PHENIX paper(s) that uses the same method: PPG243
- If the method is used in previously approved preliminary results, please give the information on the approval, i.e., when approved and some of the approved plot:
 - ▶ Au+Au direct photons using the external conversion method
 - ▶ Preliminary granted to Wenqing Fan on 2019-10-28
 - ▶ Supporting analysis note: an1417
- In both cases, please show that this established method worked well in the current data set: the Cu+Au R_γ and direct photon spectrum are consistent with the Au+Au 40–50% result from PPG243, which has comparable charged-particle multiplicity
- If you modified an established method, please highlight the modification you made: none



Outline of the analysis method

The direct photon double ratio is measured as

$$R_\gamma = \frac{\gamma^{\text{inclusive}}}{\gamma^{\text{decay}}} \approx \frac{\langle \varepsilon_\gamma f \rangle (N_{\text{incl}}/N_{\text{tag}})_{\text{data}}}{(\gamma^h/\gamma^{\pi^0})_{\text{MC}}}.$$

- ▶ N_{incl} is the inclusive conversion-photon yield from the $M_{e^+e^-}$ distribution.
- ▶ N_{tag} is obtained by combining a conversion photon with an EMCal photon and extracting the π^0 peak in $M_{e^+e^-\gamma}$.
- ▶ $\langle \varepsilon_\gamma f \rangle$ is the probability to tag the partner photon from a π^0 decay, evaluated with simulation and occupancy correction.
- ▶ γ^h/γ^{π^0} is the decay-photon cocktail correction from HELIOS.

Extraction of N_{incl} and N_{tag}

Inclusive conversion yield

- ▶ Reconstruct external conversions, $\gamma \rightarrow e^+e^-$.
- ▶ Extract N_{incl} from the $M_{e^+e^-}$ distribution after background subtraction.
- ▶ The conversion mass shape is shifted because the tracks are reconstructed assuming the event vertex, while the conversion occurs in detector material at finite radius.

Tagged π^0 yield

- ▶ Combine the conversion photon with EMCal photon candidates.
- ▶ Extract the π^0 signal from the $M_{e^+e^-\gamma}$ distribution.
- ▶ Mixed-event background is normalized outside the π^0 peak and subtracted.
- ▶ Residual background is described by a polynomial under the π^0 peak.

The nominal π^0 extraction uses a Gaussian signal plus a second-order polynomial residual background. Variations of the integration window, fit range, and polynomial order are included in the systematic uncertainty.

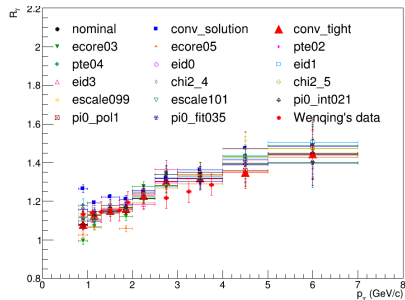
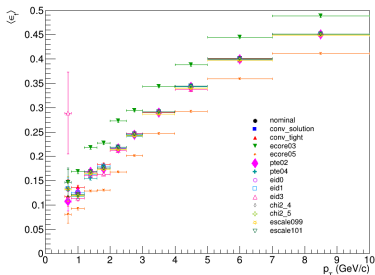
Systematic uncertainties

Systematic uncertainties are evaluated by repeating the full extraction for physically motivated cut and fit variations. Variations are grouped by source:

- ▶ conversion-pair selection;
- ▶ electron identification;
- ▶ electron-track momentum threshold;
- ▶ EMCal photon E_{core} threshold;
- ▶ EMCal shower-shape requirement;
- ▶ photon energy scale;
- ▶ π^0 peak extraction.

The systematic uncertainty is evaluated group-by-group from the spread of the variation relative to the nominal result. The final band uses asymmetric up/down uncertainties with a baseline lower bound to avoid underestimates from statistical fluctuations in individual variation points.

Efficiency and R_γ for all cuts



- There are clear trends in the efficiency as a function of the applied cuts.
- R_γ shows no dependence on the applied cuts and counting windows.
- R_γ agrees with 40-50% Au+Au results at similar multiplicity to minimum bias Cu+Au.

Analysis Issues

What were the main issues of the analysis?

- The earlier R_γ result was too flat as a function of p_T
- At high p_T , the simulated π^0 peak was too narrow because of the simulation vertex treatment, which caused a loss of tagging efficiency
- At low p_T , the $n_{cluster} == 0$ rejection removed inclusive conversion photons without a corresponding tagged photon contribution
- This made $\langle \varepsilon_\gamma f \rangle = N_{\text{tag}}/N_{\text{incl}}$ too large at low p_T

What concerns were raised in PWG meetings?

- Shape of R_γ compared to the Au+Au reference
- Treatment of systematic uncertainties

What was the resolution for these issues?

- Corrected the simulation vertex treatment, improving the high- p_T π^0 peak and tagging efficiency
- Removed the $n_{cluster} == 0$ rejection to avoid losing inclusive photons at low p_T
- Re-evaluated N_{incl} , N_{tag} , $\langle \varepsilon_\gamma f \rangle$, R_γ , and the direct photon spectrum
- Estimated systematics by varying conversion cuts, electron ID, photon cuts, energy scale, and π^0 extraction settings

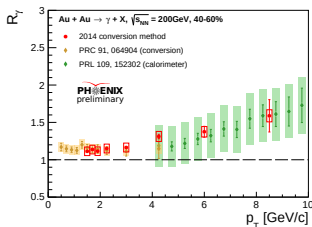
Relations to Previous Analyses

Previously published PHENIX results for the same observable:

- No published PHENIX Cu+Au direct photon result with the external conversion method

Previously published PHENIX results with a similar observable and method:

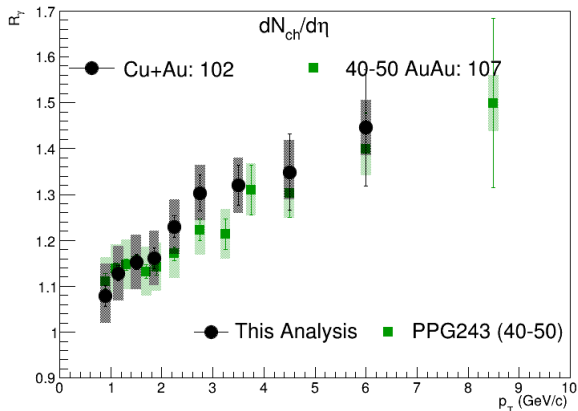
- PHENIX Au+Au direct photon measurement using the external conversion method



Previously published non-PHENIX results for similar observables:

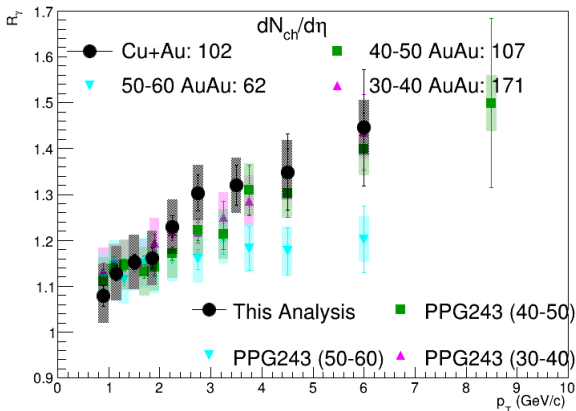
- Yes, STAR and ALICE have published related direct photon measurements

Plots Requested for PHENIX Preliminary: R_γ



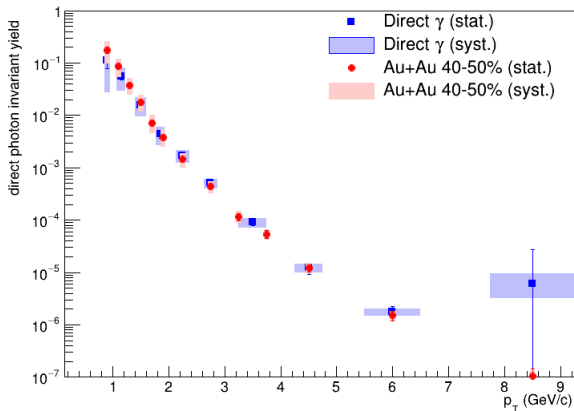
Cu+Au minimum-bias R_γ compared with the PHENIX Au+Au 40–50% reference from PPG243.

Plots Requested for PHENIX Preliminary: R_γ multiplicity comparison



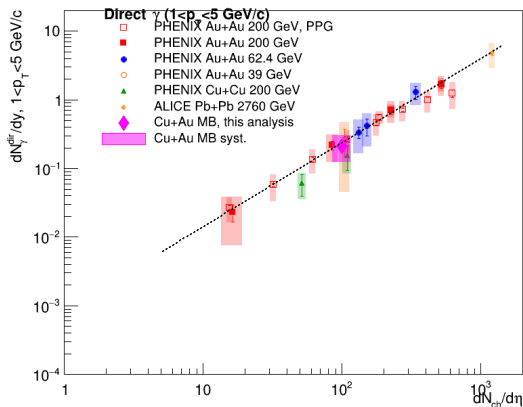
Cu+Au minimum-bias R_γ compared with PHENIX Au+Au 50–60%, 40–50%, and 30–40% reference measurements from PPG243.

Plots Requested for PHENIX Preliminary: direct photon spectrum



Direct photon invariant yield in Cu+Au minimum-bias collisions compared with the PHENIX Au+Au 40–50% reference.

Plots Requested for PHENIX Preliminary: integrated yield



Integrated direct photon yield $dN_{\gamma}^{\text{dir}}/dy$ for $1 < p_T < 5$ GeV/c versus charged-particle multiplicity at midrapidity.

Physics message

- ▶ The measured Cu+Au minimum-bias R_γ is consistent with the PHENIX Au+Au reference measurements at similar charged-particle multiplicity.
- ▶ The extracted direct photon invariant yield agrees with the Au+Au 40–50% result within uncertainties.
- ▶ The Cu+Au integrated direct photon yield follows the global scaling of $dN_\gamma^{\text{dir}}/dy$ with $dN_{\text{ch}}/d\eta$ observed across collision systems and beam energies.
- ▶ The result extends the PHENIX external-conversion direct photon program to the asymmetric Cu+Au system.

The Path to the Final Results

What improvements are expected for the final results?

Using whole available data and centrality bins.

Is all data available used?

No, about 30% of the Run-12 Cu+Au minimum-bias sample is used.

How much statistical/systematical gain is expected before final publication?

About $\sqrt{3}$ statistical gain is expected.

Provide a detailed timeline to publication:

Prepare article draft by fall 2026.

Provide details on who will do the work and milestones towards final results:

Tongzhou Guo, A. Drees, and Yu. Mitrankov.